

- Cell junctions
 in local cell signaling, 215f
 plasmodesmata in plants, 119f, 120
 tight junctions, desmosomes, and gap junctions in animals, 120f
- Cell-mediated immune response, 957f, **963**, 966f–967f
- Cell migration
 in organogenesis, 1054f, 1055
- Cell motility, 112, 113t, 114, 115f–117f, 1056f
- Cell plate, **241**, 242f–243f
- Cells, **5f**, **759**
 animal. *See* Animal cells
 auxin in differentiation of, 850
 auxin in elongation of, 848, 849f
 blood, 934f–936f, 937
 calculating volume and surface area of, 99
 cell fractionation in study of, 96f, 97
 cellular integration of, 121f
 cellular membranes of. *See* Cellular membranes
 cellular respiration and. *See* Cellular respiration
 climate change effects on, 1280f
 communication between. *See* Cell signaling
 cytokinins in division and differentiation of, 850
 differentiation of. *See* Differentiation, cell
 division of, as fundamental to life, 235. *See also* Cell cycle; Cell division
 eukaryotic vs. prokaryotic, 97f–98f, 99. *See also* Eukaryotic cells; Prokaryotic cells
 as fundamental units of life, 5f–6f, 93f
 locations of enzymes in, 161f
 metabolism of. *See* Metabolism
 microscopy in study of, 94f–95f, 96
 molecular machinery in, 122f–123f
 photosynthesis and. *See* Photosynthesis
 plant. *See* Plant cells
 programmed death of. *See* Apoptosis
 programming of, by viral DNA, 315f–316f, 317
 protein folding in, 83f
 protocells as first, 528f
 sequential gene regulation in differentiation of, 383, 384f
 shape changes in animal morphogenesis, 1056f–1057f
 in sickle-cell disease, 502f
 size range of, 94f
 of small intestine, 125
 stem cells. *See* Stem cells
 transcription specific to type of, 374–375, 377f
 unity in, 13f
 water regulation of, 142
- Cell sap, 108
- Cell signaling. *See also* Signal transduction pathways
 by animal endocrine and nervous systems, 880f
 in apoptosis, 229f–231f
 cancer and interference with normal, 389, 390f, 391
 in cell cycle control system, 244, 245f–248f.
 See also Cell cycle control system
 cellular membrane selective permeability and, 130f
 cilia in, 114
 endocrine system, 1000f–1004f
 evolution of, 213f, 214, 233
 feedback regulation of, 1005–1006
 fight-or-flight responses in, 212f
 local and long-distance, 215f, 216
 mechanical, 119
 pathways of. *See* Endocrine signaling;
 Neuroendocrine signaling
 reception stage of, 217f–221f
 response stage of, 226f–228f, 229
 in simple endocrine pathway, 1004, 1005f
 in simple neuroendocrine pathway, 1005f
 symplastic, 801–802
 three stages of, 216f, 217
 transduction stage of, 221, 222f–225f
- Cell-surface proteins, 119f, 130f
- Cell-surface receptor tyrosine kinases (RTKs), 250
- Cell-surface transmembrane receptors, 217f–220f
- Cellular hormone response pathways, 1002f–1003f
- Cellular innate immune defenses, 954, 955f
- Cellular-level herbivore defenses, plant, 868f
- Cellular membranes. *See also* Plasma membranes
 active transport across, 136, 137f–138f, 139
 animal, 674
 bulk transport across, by exocytosis and endocytosis, 139, 140f, 141
- evolution of differences in lipid composition of, 129
 fluidity of, 128f, 129
 as fluid mosaics of lipids and proteins, 127f, 128
 interpreting scatter plots on glucose uptake across, 136
 membrane carbohydrates in cell-cell recognition by, 130f–131f
 membrane proteins of, 129f–130f, 132
 of mitochondria, 110, 111f
 movement across plant cell, 209f
 nuclear envelopes, 102, 103f
 organelles and internal, 98f, 99
 passive transport as diffusion across, 132, 133f–135f
 phospholipids in, 74f, 75
 in plant response to cold stress, 865
 selective permeability of, 127f, 131, 132f
 specialized prokaryotic, 577f
 synthesis and sidedness of, 131f
- Cellular respiration, **165**
 ATP production by catabolic pathways and, 165
 ATP yield at each stage of, 177f, 178
 bar graphs of, 179
 biosynthesis in anabolic pathways and, 183
 in carbon cycle, 1250f
 as catabolic, 144, 165
 in energy flow and chemical cycling, 9f, 164f, 165
 enzymes for, in mitochondria, 161f
 evolutionary significance of glycolysis in, 182
 fermentation vs., 165, 179, 180f–181f, 182
 glycolysis in, 170f–171f
 metabolic rate and, 890f
 mitochondria in, 109–110, 111f
 monosaccharides in, 69
 origin of, 534, 535f
 overall reaction for, 149
 oxidative phosphorylation in, 174f–177f
 oxygen diffusion and, 133
 photosynthesis vs., 190. *See also* Photosynthesis in plant cells, 209f
 pyruvate oxidation and citric acid cycle in, 171f–173f
 redox reactions and, 165, 166f–168f
 regulation of, via feedback mechanisms, 183f, 184
 scale of, 122f
 stages of, 168, 169f
 using cell fractionation to study, 97
 versatility of catabolic pathways and, 182f, 183
- Cellular slime molds, 612, 613f
- Cellulose, **71**
 microfibrils, 777f
 in plant cell walls, 70f–71f, 72, 118
 as product of photosynthesis, 206, 207f
 proteins synthesizing, 619f
 water and, 49
- Cellulose synthase, 118
- Cell walls, **118**
 cellulose in plant, 619
 fungal cell, 100f, 661
 osmosis, water balance, and, 134f–135f
 plant cell, 101f, 118f
 prokaryotic, 97f, 574, 575f
- Cenozoic era, 530f, 533f, 539f, 679
- Center for Plant Conservation, 1261
- Centipedes, 708, 709f
- Central axis, 680f
- Central canal, 1087f
- Central disk, sea star, 714f
- Central dogma, DNA, 339
- Central nervous system (CNS), **1069**, **1086**. *See also* Brains
 neuronal plasticity of, **1100f**
 neurons of, 1069
 neurotransmitters and, 1080
 peripheral nervous system and, 1086, 1087f–1090f
 in sensory systems, 1108–1109
 structure and function of vertebrate, 1087f–1088f
- Central vacuoles, 101f, **108f**
- Centrifuge, 96f, 97
- Centrioles, **114f**
- Centromeres, **236f**, 264
- Centromeric DNA, 452
- Centrosomes, 100f, **114f**, 238f, **240f**
- Centrostephanus rodgersii*, 1281f
- Century plants, 1200, 1201f
- Cephalization, 1086–1087
- Cephalochordata (lancelets), 719f, **720**, 721f–722f
- Cephalopods, 699, 701f, 702
- Ceratotherium simum cottoni*, 1266
- Cercozoans, **608f**, 609
- Cerebellum, 1091f, **1092f**–1093f
- Cerebral cortex, **1093f**, 1096, 1097f–1099f, 1100, 1114
- Cerebral ganglia, 704f–710f
- Cerebral hemispheres, 1093f
- Cerebrospinal fluid, 946f, 1087f
- Cerebrum, 1091f, **1092f**–1093f, 1094
- Certainty of paternity, 1150, 1151f
- Cervical cancer, 974f
- Cervix, **1026**, 1027f
- Cetaceans, 481f–482f, 745f
- Cetartiodactyla, 745f
- cGMP (cyclic GMP), 224, 844f, 845
- Chaetae, 703, 704f
- Chagas' disease, 600
- Chain worms, 694
- Chambered nautilus, 701f, 702
- Chameleons, 673f, 735f
- Chamerion angustifolium*, 636f
- Chamois (*Rupicapra rupicapra*), 901f
- Change, global. *See* Global change
- Change, spontaneous, 148f
- Channel proteins, 132, 135f, 136
- Chaparral biomes, **1174f**
- Character displacement, **1216**, 1217f, 1237
- Characters, **270**
 dominant vs. recessive traits and, 271f, 272t
 multifactorial, 282f
 shared ancestral and shared derived, 560, 561f
 taxonomy and, 554
 traits and, 270–271
- Character tables, 561f
- Charadrius melodus*, 1190f
- Chargaff, Edwin, 317f, 318
- Chargaff's rules, 317f, 318, 320
- Charged tRNA, 349f
- Charophytes, 610, 619f–620f
- Charpentier, Emmanuelle, 426
- Chase, Martha, 316f, 317
- Cheating behavior, 1159
- Checkpoints, cell cycle control system, **245f**–247f
- Cheetahs, 212f, 1204f
- Cheliceræ, **708**
- Chelicerates, **707**, 708f
- Chemical bonds, **36**
 with carbon, 56, 58, 59f, 60
 covalent, 36f–37f
 hydrogen bonds, 38, **39f**
 ionic, 37, 38f
 van der Waals interactions, 39
- Chemical cycling
 in ecosystems, 1238f–1240f, 1241
- Chemical cycling, energy flow and. *See* Biogeochemical cycles; Energy flow and chemical cycling
- Chemical defense, prey, 1218f
- Chemical digestion, 907f–908f, 909
- Chemical energy, 143–**144**, 145f, 889, 890f. *See also* Cellular respiration; Photosynthesis
- Chemical equilibrium, **41**
 buffers and, 52–53
 in chemical reactions, 41
 free energy change and, 147, 148f
 metabolism and, 148, 149f–150f
- Chemical mutagens, 360
- Chemical reactions, **40**
 activation energy barrier of, 153, 154f–155f
 chemical energy in, 144, 145f
 endergonic, 148, 149f
 enzymatic catalysis of. *See* Enzymatic catalysis
 exergonic, 148, 149f, 154f, 165
 free energy change and, 147, 148f
 making and breaking of chemical bonds by, 40f–41f
 metabolism and. *See* Metabolism
 in photosynthesis, 190f, 191. *See also* Light reactions
- Chemical signals, 1068. *See also* Animal hormones; Hormones; Plant hormones
- Chemical structure, DNA, 319f
- Chemical synapses, 1077, 1078f. *See also* Synapses
- Chemical work, 150–151, 152f
- Chemiosmosis, 169, 175, **176f**–177f, 191f, 199f–200f, 201
- Chemistry
 atoms in, 30f–36f
 biological molecules in. *See* Biological molecules